BOARD OF SECONDARY EDUCATION, KARACHI

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

BIOLOGY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

SECTION "A" MULTIPLE CHOICE QUESTIONS (MCQs)

Maximum Marks: 24

Note: NOTE: (i) Attempt all the Questions of this Section. (ii) Do not copy down the part questions. Write only the answer against the proper number of the Question and its part according to the question Paper. (iii) Each question carries 1 Mark.

Q1. Choose the correct Answer for each from the given question: (For Fresh Candidates of 2022)

Q1. The biological function which performs gaseous exchange

(A) Photosynthesis
(B) Respiration
(C) Growth
(D) Alveoli

Q2. The network of blood capillaries present in the layer of skin:

(A) Epidermis
(B) Dermis
(C) Hypodermis
(D) Endodermis

Q3. The harmone ADH release from:

(A) Pituitary gland
(B) Kidneys
(C) Liver
(D) Lungs

Q4. The automatic process of altering focus to get sharper image of near object is:

(A) Vision
(B) Accommodation
(C) Focus
(D) All of these

Q5. "Kitab ul Manazir" was written by:

(A) Jabir Bin Hayan
(B) Ibn al Haitham
(C) Ali Ibn Isa
(D) Bu Ali Sina

Q6. The framework which gives shape to any structure is:

(A) Bone
(B) Skeleton
(C) Cartilage
(D) Architecture

Q7. The part of brain involved in reasoning is:

(A) Fore brain
(B) Cerebrum
(C) Cortex
(D) Frontal lobe

Q8. The pair of muscles where both work in opposite direction is:

(A) Antagonist
(B) Cardiac
(C) Smooth
(D) Abductor

Q9. The type of seed production without fusion of male and female gametes is:

(A) Parthenocarpy
(B) Apomixes
(C) Grafting
(D) Scion or Crystal

Q10. The way by which gene transmits characters from parents to offspring is:

(A) Genetics
(B) Inheritance
(C) Heredity
(D) Allele

Q11. Any biological region recognized by its climate or vegetation is:

(A) Ecosystem
(B) Biomes
(C) Biosphere
(D) Biomass or Sub class

Q12. The study of complete graph of human genome is called:

(A) PCR
(B) HGP
(C) Medicine
(D) Soma-Clonal

Q13. The bread dough rises during alcoholic fermentation is due to:

(A) Methyl alcohol
(B) CO2
(C) Ethyl alcohol
(D) H2O

Q14. Father of antiseptic is:

(A) Alexander Fleming
(B) Edward Jenner
(C) Joseph Lister
(D) Oswald Schmiedeberg

Q15. Drugs that slow normal brain functioning, is called:

(A) Narcotics
(B) Hallucinogen
(C) Marijuana
(D) Sedatives

Q16. Disorder which is associated with degeneration of alveoli is:

(A) Bronchitis
(B) Lungs cancer
(C) Asthma
(D) Emphysema

Q17. The outer part of DNA helix made up of sugar and phosphate, is called:

(A) Nucleoprotein
(B) Upright
(C) Rungs
(D) Phosphoester

Q18. The vitamin necessary for proper vision is:

(A) Vitamin A
(B) Vitamin B
(C) Vitamin C
(D) Vitamin D

Q19. The large muscle of arm is:

(A) Bicep
(B) Tricep
(C) Tetracep
(D) Pentacep

Q20. The female gametes are fertilized in the rear end of:

(A) Oviduct
(B) Fallopian tube
(C) Ovaries
(D) All of these

Q21. Plants grow near coastal area are called:

(A) Xerophyte
(B) Halophyte
(C) Epiphyte
(D) Hygrophyte

Q22. Each stoma is formed by:

(A) One guard cell
(B) Two guard cells
(C) Three guard cells
(D) Four guard cells

Q23. The cold war between the organisms of a community occupying the same habitat in ecosystem, is called:

(A) predation
(B) competition
(C) mutualism
(D) commensalism

Q24. The part of brain involved in balance and precision in movement is:

(A) Cerebrum
(B) Cerebellum
(C) Thalamus
(D) Medulla Oblongata

SECTION “B” (Short Answer Question)

Maximum Marks: 24

> Note: NOTE: Attempt any EIGHT questions from this section. Each question carries Three (3) marks.

Q2. Write three differences between plant cell and animal cell.

Q3. Draw neat and labelled diagram of Ovule.

Q4. How does light affect on ecosystem?

Q5. What is rickets? Write its causes.

Q6. How is the asthma patient characterized?

Q7. Why is skin considered as excretory organ?

Q8. Define following terms:

(a) Homologous chromosome

(b) Allele

(c) Recessive Allele

Q9. What is algal bloom and how it destroys the life of an aquatic ecosystem?

Q10. Draw neat and labelled diagram of hinge OR ball and socket joint.

Q11. Why do antibiotics not affect against viral infection?

Q12. Distinguish between sexual and asexual reproduction.

Q13. State Mendel's law of independent assortment. (No need to draw checkerboard)

Section C (Detailed –Answer Question)

Maximum Marks: 12

> Note: NOTE: Attempt any TWO questions from this Section. Each question carries SIX (6) marks.

Q14. Draw neat and labelled diagram of human ear.

Q15. Describe the urinary system of man with the help of diagram.

Q16. What is germination? Write the necessary conditions of it and also describe the epigeal and hypogeal germination.